

Abstract

Virtual try-on methods based on latent diffusion face two fundamental challenges: 1) warping garment details in latent space introduces distortion, making it difficult to preserve colors, logos, patterns, and textures, and 2) without true try-on triplets for training (images of the same person wearing different garments in identical poses), the field has adopted masking approaches that erase body-specific characteristics and constrain transformation volume. FASHN VTON v1.5 addresses both challenges with three key properties: 1) pixel-space generation that operates directly on RGB pixels without VAE encoding, preserving garment details, 2) maskless inference that generates try-on results without masking, preserving body identity and removing volume constraints, and 3) interactive performance with a 972M parameter architecture optimized for ~5 second inference on H100 GPUs. FASHN VTON v1.5 demonstrates that pixel-space generation combined with maskless inference enables high-fidelity virtual try-on suitable for consumer-facing applications. Released under Apache 2.0, FASHN VTON v1.5 is the first permissive open-source virtual try-on model and one of few pixel-space diffusion architectures publicly available for research and commercial use.

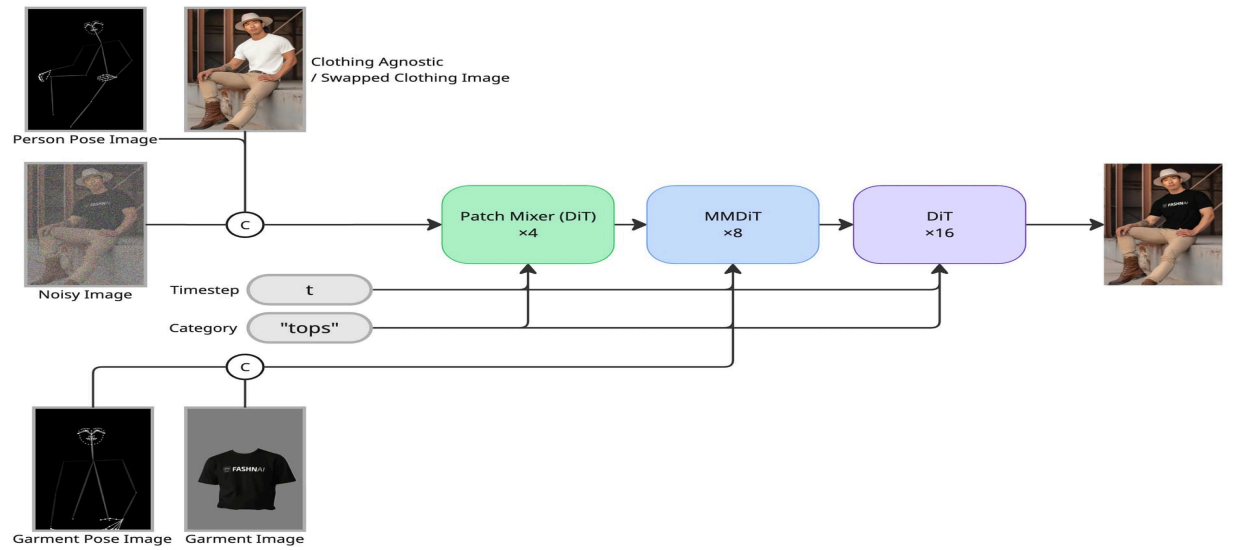
Method

Architecture

FASHN VTON v1.5 uses an MMDiT [1] architecture that processes person and garment images jointly. 8 double-stream blocks handle cross-modal attention between person and garment tokens, followed by 16 single-stream blocks for self-attention over the concatenated sequence. 4 patch-mixer blocks [2] pre-process image patches before the main transformer.

The model operates directly on RGB pixels instead of compressed latent space. The only spatial reduction is a 12×12 patch embedding. Pose keypoints are provided as single-channel grayscale images concatenated to the person and garment inputs.

Category embeddings ("tops", "bottoms", "one-pieces") are added to timestep conditioning.



Training

FASHN VTON v1.5 was trained from scratch in two phases. Phase 1 trained on 18M masked try-on pairs. Phase 2 trained on a 50/50 mix of masked pairs and 4M synthetic triplets generated by the Phase 1 checkpoint. These triplets consist of people wearing alternative garments paired with their originals as ground truth, enabling the model to perform try-on from an unmasked input while preserving body shape.

Training from scratch in pixel space is computationally demanding. The patch-mixer blocks addressed this by allowing up to 75% of tokens to be dropped during training, significantly reducing memory and compute requirements.

Results

The key advantage of maskless inference is freedom from the constraints imposed by segmentation masks. The examples below compare v1.5 against FASHN VTON v1, the previous state-of-the-art which relied on masked inference. They highlight challenging scenarios: voluminous garments that exceed the original outfit's boundaries, hair regions where the original garment leaks through due to segmentation errors, tattoos

and body characteristics that masked methods typically erase, and cultural garments like hijabs that require preservation during swaps.

Limitations

Resolution: Output is limited to 576×864 compared to 1K+ resolutions achievable with VAE-based architectures.

Body preservation: Maskless inference significantly improves body preservation compared to masked methods, but is not yet perfect. Since synthetic triplets were generated by a masked model, some body characteristics may still shift slightly.

Garment removal: In maskless mode, the original garment may not be fully removed, especially in long-to-short transitions or bulky-to-slim conversions. The model preserves body shape but may leave traces of the original clothing.